

Jaesin Ahn

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RESEARCH INTERESTS

AI Safety, Trustworthy AI, Generative AI, Machine Unlearning, Computer Vision

PUBLICATIONS

[C1] **Jaesin Ahn** and Heechul Jung, "Mitigating Sexual Content Generation via Embedding Distortion in Text-conditioned Diffusion Models.", **The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)**.

[J1] **Jaesin Ahn** and Heechul Jung, "A Self-Attention Classifier Head for Improved Image Classification and Interpretability of ViT," **Electronics Letters**, vol. 61, no. 1, 2025.

[J2] **Jaesin Ahn** and Heechul Jung, "GDoT: A Gated Dual Domain Transformer for Enhanced MRI Off-Resonance Correction," **Neurocomputing**, 129918, 2025.

[J3] **Jaesin Ahn**, Jiuk Hong, Jeongwoo Ju, and Heechul Jung, "Redesigning Embedding Layers for Queries, Keys, and Values in Cross-Covariance Image Transformers," **Mathematics**, vol. 11, no. 8, p. 1933, Apr. 2023.

[C2] **Jaesin Ahn**, Hyun-Joo Lee, Inchul Choi, Minho Lee, "Skip-StyleGAN: Skip-Connected Generative Adversarial Networks for Generating 3D Rendered Image of Hand Bone Complex.", **23rd International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2020)**, Virtual

[J4] Sanjar Karshiev, Faisal Saeed, **Jaesin Ahn**, Abdul Rehman, Muhammad Diyan, Shrooq Alsenan, Heechul Jung, "LEAF: A Lightweight Language-Enhanced Model for Forestry Analysis in Remote Sensing Imagery.", **IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing**, 2026.

[C3] Doyoung Kwon, **Jaesin Ahn**, Jaeil Kim, Inchul Choi, Sungmoon Jeong, Young-Sup Lee, Jaechan Park, Minho Lee, "Siamese U-Net with Healthy Template for Accurate Segmentation of Intracranial Hemorrhage.", **22nd International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2019)**, Shenzhen, China

[Under Review] "Diagnosing LLM De-Identification Failure in Clinical Text via Mechanistic Interpretability"

[Under Review] "Ontology-Guided Semantic Mixing with Adaptive LoRA Experts for Long-Tailed Remote Sensing Recognition"

[Under Review] "When Metadata Matters: Impact of Recording and Demographic Factors on Respiratory Sound Classification"

CHALLENGES & AWARDS

2025 AI CHAMPION Challenge

- 7th place (Top 1.11%), Recipient of the IITP President's Award
- Hosted by Ministry of Science and ICT (MSIT)

2023 NeurIPS Machine Unlearning Challenge

- 8th place (Top 0.6%)
- Organized by Google DeepMind

EDUCATION

Kyungpook National University

Daegu, Korea

Department of Artificial Intelligence

Mar. 2021 – Feb. 2026

Ph.D. in Artificial Intelligence. GPA: 4.23/4.3

- Thesis title: Refining Embeddings in Transformers: Image Classification, MRI Reconstruction, and Safe Image Generation

Kyungpook National University

Daegu, Korea

School of Electronic and Electrical Engineering

Mar. 2019 – Feb. 2021

M.S. in Electronic and Electrical Engineering. GPA: 4.12/4.3

- Thesis title: Neighborhood Skip-StyleGAN for Generating 3D Hand Bone Complex from a Single X-ray Image

Kyungpook National University

Daegu, Korea

School of Electronics Engineering

Mar. 2013 – Feb. 2019

B.S. in Electronics Engineering. GPA: 3.82/4.3

TECHNICAL SKILLS

Languages: Python, Bash/Shell Scripting

Deep Learning: PyTorch, Hugging Face

Research: Mechanistic Interpretability (TransformerLens, NNsight)

Tools: Docker, Git/GitHub, Linux/Ubuntu, Claude Code

GLOBAL RESEARCH EXPERIENCES

Graduate Research Assistant, *Memorial Sloan Kettering Cancer Center (MSKCC), New York, USA* (Apr. 2022 - Mar. 2023)

- Project: Research on Unsupervised Domain Adaptation Technology Based on Deep Learning (Supported by IITP grant)
- Developed a **transformer-based correction network** to mitigate off-resonance **artifacts caused by accelerated MRI scanning** with non-Cartesian sparse trajectories.
- This research led to the publication of the "GDoT" paper in *Neurocomputing*.

PROJECT EXPERIENCES (SELECTED)

Developing the Next-Generation General AI with Reliability, Ethics, and Adaptability, IITP, Korea (Apr. 2025 - Current)

- **Designing a reliable AGI architecture** by analyzing stability issues in **multi-agent interactions** and exploring **model routing** strategies for efficient inference.
- Focusing on enhancing model capabilities and adaptability across diverse specialized domains, including medical and mathematical reasoning.

Evaluation of Privacy Risk Mitigation Technologies for Generative AI, Personal Information Protection Commission, Korea (May. 2024 - Oct. 2024)

- **Analyzed privacy leakage risks** specifically in **Korean Large Language Models (LLMs)** and evaluated the effectiveness of mitigation strategies.
- Verified the performance of privacy-preserving techniques, including deduplication, filtering, differential privacy, and machine unlearning.

PATENTS & SOFTWARES

Heechul Jung, **Jaesin Ahn**, "Mechanistic Interpretability-based Backdoor Detection and Neutralization Apparatus and Method for Large Language Models", KR Patent 10-2025-0112676, filed Aug. 13, 2025

Heechul Jung, **Jaesin Ahn**, "Image Classification Apparatus, Attention-based Classifier Head Thereof, and Image Classification Method Using the Same", KR Patent 10-2025-0111311, filed Aug. 12, 2025

Heechul Jung, **Jaesin Ahn**, "Apparatus for Content Generation and Learning Method Thereof", KR Patent 10-2025-0044952, filed Apr. 7, 2025

Heechul Jung, **Jaesin Ahn**, Soyoun Bang, "Incremental learning device and incremental learning method", KR Patent 10-2025-0044706, filed Apr. 7, 2025

Heechul Jung, **Jaesin Ahn**, "Apparatus and Method for restoring a 3D Magnetic Resonance Imaging Including Blurring into a Clean Magnetic Resonance Image with the Blurring Removed", KR Patent 10-2024-0149695, filed Jan. 23, 2025

Heechul Jung, **Jaesin Ahn**, "Query, Key, Value Embedding Technique Using Non-linearity and Shared Features, KR Patent 10-2023-0096881, filed Jul. 25, 2023

Minho Lee, **Jaesin Ahn**, " Method for Generating Rotated Hand Bone 2D Projection Image from 1 Hand Bone 2D Projection Image, Recording Medium and Device for Performing the Method", KR Patent 10-2343363-0000, issued Dec. 21, 2021